Solid State Physics — Week 3 Assignment Questions & Answers

Question 21

Find out the linear density of FCC crystal in the direction [100] in terms of atomic radius R.

(a) $\frac{1}{2R}$

(b) $\frac{1}{\sqrt{2}R}$

(c) $\frac{1}{2\sqrt{2}R}$

(d) $\frac{1}{3\sqrt{2}R}$

Accepted answer: (c)

Question 22

Find the linear density for FCC crystal in the [111] direction in terms of the atomic radius R.

(a) $\frac{1}{2\sqrt{6}R}$

(b) $\frac{1}{3\sqrt{2}R}$

(c) $\frac{1}{\sqrt{2}R}$

(d) $\frac{1}{R}$

Accepted answer: (a)

Question 23

Calculate the values of linear density along the [111] direction vectors for the FCC copper. Consider the atomic radius of copper is 0.128 nm.

(a) $1.21 \times 10^{7}\ m^{-1}$

(b) $1.39 \times 10^{8}\ m^{-1}$

(c) $1.59 \times 10^{9}\ m^{-1}$

(d) $1.67 \times 10^{12}\ m^{-1}$

Accepted answer: (c)

Question 24

Find out the linear density of BCC crystal in the [110] direction in terms of the atomic radius R. consider the lattice parameter of the crystal is a.

(a) $\frac{\sqrt{3}}{4\sqrt{2}\,R}$

(b) $\frac{\sqrt{3}}{R}$

(c) $\frac{2\sqrt{3}}{R}$

(d) $\frac{3\sqrt{3}}{R}$

Accepted answer: (a)

Question 25

Calculate the linear density for BCC crystal in the [111] direction in terms of the atomic radius R. Consider the lattice parameter of the BCC crystal is a.

(a) $\frac{1}{2R}$

(b) $\frac{1}{3R}$

(c) $\frac{1}{4R}$

(d) $\frac{1}{5R}$

Accepted answer: (a)

Question 26

Calculate the values of linear density along the [111] direction vectors for the BCC iron. Consider the atomic radius of iron is 0.124 nm.

(a) $2.11 \times 10^{7}\ m^{-1}$

(b) $4.03 \times 10^{9}\ m^{-1}$

(c) $5.12 \times 10^{11}\ m^{-1}$

(d) $6.67 \times 10^{15}\ m^{-1}$

Accepted answer: (b)

Question 27

Calculate the planar density of FCC crystal for (111) planes in terms of the atomic radius R. Consider the lattice parameter of the cubic crystal is a.

(a) $\frac{1}{2\sqrt{3}R^2}$

(b) $\frac{1}{\sqrt{2}R^2}$

(c) $\frac{1}{3R^2}$

(d) $\frac{1}{4R^2}$

Accepted answer: (a)

Question 28

Calculate the values of planar density of (100) plane for the FCC aluminium. Consider the atomic radius of aluminium is 0.143 nm.

(a) $1.22 \times 10^{19}\ m^{-2}$

(b) $2.47 \times 10^{21}\ m^{-2}$

(c) $4.43 \times 10^{25}\ m^{-2}$

(d) $5.12 \times 10^{30}\ m^{-2}$

Accepted answer: (a)

Question 29

A crystal plane intercepts crystal axes at 0.5a, b, and is parallel to the c axis. The miller indices of the plane is

(a) (120)

(b) (210)

(c) (021)

(d) (102)

Accepted answer: (b)

Question 30

Which of the following structure have the highest packing factor?

(a) $FCC\ crystal$

(b) $Simple\ cubic\ crystal$

(c) $BCC\ crystal$

(d) $Diamond\ structure\ (DC)$

Accepted answer: (a)